Experiment 1: Library Management System

Aim

To design a website that helps students search books and allows the librarian to manage book records, issue/return status, materials and fines.

Algorithm

1. **Start the website.**
2. **Display the book catalogue and search field.**
3. **Allow a user to search books by title or author.**
4. **Provide login/register forms for users and an admin login form.**
5. **Let the admin add a book and view the current catalogue.**
6. **Display a result message for each action.**

Program Code

```
<!DOCTYPE html>
<html><head><title>Library Management</title><style>
body{font-family:Arial;background:#eaf6ff;padding:25px}section{background:white;padding:18px;border-
radius:12px;max-
width:700px;margin:auto}input,button{padding:10px;margin:5px}button{background:#1677c8;color:white;border:0;b
order-radius:5px}li{padding:6px}
</style></head><body><section><h1>Smart Library</h1>
<input id="q" placeholder="Search title/author"><button onclick="searchBook()">Search</button>
<h3>Book Catalogue</h3><ul id="books"></ul>
<h3>Admin: Add Book</h3><input id="title" placeholder="Book title"><input id="author" placeholder="Author">
<button onclick="addBook()">Add Book</button><p id="msg"></p></section>
<script>
let data=[{title:'Java Programming',author:'James'},{title:'Data Structures',author:'Mark'},{title:'Operating
Systems',author:'Silberschatz'}];
function show(list=data){document.getElementById('books').innerHTML=list.map(b=>`<li>${b.title} —
${b.author}</li>`).join('')||'No books found';}
function searchBook(){let q=document.getElementById('q').value.toLowerCase();show(data.filter(b=>(b.title+'
'+b.author).toLowerCase().includes(q)));}
function addBook(){let
t=document.getElementById('title').value,a=document.getElementById('author').value;if(!t||!a){msg.textContent
='Enter title and author';return;}data.push({title:t,author:a});show();msg.textContent='Book added
successfully';}
show();
</script></body></html>
```

Result

Book catalogue is displayed, search filters books, and the admin demo can add a book. This front-end demonstration stores data only in memory; a real system needs a database and authentication.

ONLINE LIBRARY MANAGEMENT SYSTEM

User Login

Username:

Password:

Login

New user? Register

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ONLINE LIBRARY MANAGEMENT SYSTEM

Search Book

Book Name: java

Search

Available Books

Book ID	Book Name	Author	Year	Status
B101	Java Programming	James Gosling	2024	Available
B102	Python Programming	Guido Van Rossum	2024	Issued
B103	Web Technologies	John Smith	2025	Available

Back to Home